Notes_25.13: What is Cross Validation?

Python 3.10+

Jupyter Notebook

PDF notes.pdf

🌐 **Language: English** | हिंदी / Hinglish | ⬅ Lecture 25 Overview

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📌 Executive Summary

K-Fold Cross Validation, Stratified K-Fold for class imbalance, and variance reduction.

🎯 Mathematical Formulations

K-Fold Cross Validation Estimator:

$$ \text{CV}*K* = *frac{1}{K}* *sum*{k=1}^K \text{Score}_k $$

Stratified Fold Class Ratio Invariant:

$$\frac{N_{k,+}}{N_k} \approx \frac{N_+}{N} \quad \forall; k \in 1, 2, \ldots, K$$

💡 Conceptual Principles & Architecture

Single Train-Test Split Ki Kamzori:

Agar hum sirf ek baar train/test split karte hain, toh ho sakta hai ki test set mein coincidental easy ya tough examples chale jayein. Isse performance estimate unstable ho jata hai.

K-Fold Solution:

Dataset ko K equal hisso (folds) mein divide kiya jata hai. Har iteration mein 1 fold test set banta hai aur baki $K-1$ folds train set bante hain. Yeh process K baar chalta hai aur sabhi scores ka average nikala jata hai!

- **Stratified K-Fold:** Imbalanced classification mein use hota hai taaki har fold mein positive aur negative class ka ratio original dataset jaisa hi barabar rahe.

💻 Python Production Implementation

```
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
import pandas as pd

df = pd.read_csv("heart.csv")
X = df.drop("target", axis=1)
y = df["target"]

pipe = make_pipeline(StandardScaler(), KNeighborsClassifier(n_neighbors=5))
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

scores = cross_val_score(pipe, X, y, cv=skf, scoring='accuracy')
print("5-Fold Scores:", scores.round(4))
print(f"Mean CV Accuracy: {scores.mean():.4f} (+/- {scores.std():.4f})")
```